



Josue Valenzuela Perez

Date of birth: 01/12/2004 | **Place of birth:** Celaya, Mexico | **Nationality:** Mexican | **Gender:** Male |

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josuevp-portfolio.vercel.app | **Website:** <https://github.com/JosueVP17> | **LinkedIn:** [Josue Valenzuela Perez](#) |

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ABOUT ME

Computer Systems Engineering student specializing in Machine Learning and Deep Learning, with research and applied experience developing end-to-end AI systems. Experienced in model development, optimization, and ML pipelines. Academic experience at Universidad de Guanajuato and Chalmers University of Technology, complemented by research and award-winning applied AI projects.

EDUCATION AND TRAINING

22/08/2022 – CURRENT Salamanca, Mexico

BACHELOR OF COMPUTER SYSTEMS ENGINEERING Universidad de Guanajuato

Computer Science and Software Engineering, with emphasis on Artificial Intelligence, Machine Learning, Data Science, Computer Vision, Algorithms and Data Structures, Database Systems, Software Development, Computer Networks, Operating Systems, and Web Technologies. The program provides training in the design, development, and evaluation of computational solutions, including the application of machine learning and deep learning methods to real-world problems.

Website <https://www.ugto.mx> | **Field of study** Information and Communication Technologies (ICTs) not further defined |

Level in EQF EQF level 6

18/01/2026 – 01/06/2026 Gothenburg, Sweden

EXCHANGE SEMESTER Chalmers University of Technology

Artificial Intelligence, Machine Learning, Data Science, Large-Scale Data Processing, Optimization, and AI Systems Design.

Website <https://www.chalmers.se> | **Field of study** Information and Communication Technologies (ICTs) not further defined |

Level in EQF EQF level 6

WORK EXPERIENCE

AI & DATA SCIENCE RESEARCH INTERN – LABORATORIO NACIONAL CONAHCYT EN INTELIGENCIA ARTIFICIAL Y CIENCIA DE DATOS (LNC-IACD) – 20/08/2026 – Current – ECATEPEC DE MORELOS, MEXICO

Conducting research and technological development activities in Artificial Intelligence and Data Science. Applying machine learning and deep learning methods to research problems, including data processing, model development, experimentation, and evaluation. Participating in research projects within the LNC-IACD.

UNIVERSITY RESEARCH ASSISTANT – UNIVERSIDAD DE GUANAJUATO – 28/09/2025 – 28/05/2026 – SALAMANCA, MEXICO

Developed a computer vision and signal processing system for non-contact heart rate estimation from hand videos. Implemented methods for hand region detection, extraction and processing of photoplethysmographic (PPG) signals, signal filtering, and heart rate estimation. Conducted experiments to evaluate the accuracy and robustness of the proposed approach.

HONOURS AND AWARDS

09/07/2026

1st Place – Video Analytics for Security, National Competition – Tenaxes & Seguridad Privada Okip S.A. de C.V.

Awarded first place in Category C (Video Analytics for Security) for Sentinel CV, an artificial intelligence system developed for automated video analysis in security applications. The project applied computer vision and deep learning techniques to detect and analyze relevant events in video footage.

Awarded third place in a competitive programming competition organized by the ACM Student Chapter. The competition involved solving algorithmic and programming problems under time constraints, requiring skills in algorithm design, problem-solving, and computational thinking.

22/08/2022

“Chavos que Inspiran” Scholarship Recipient – BBVA México

Selected as a recipient of the “Chavos que Inspiran” scholarship, a BBVA México program supporting high-achieving students in Mexico throughout their entire university studies.

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CONFERENCES AND SEMINARS

22/10/2025 – 24/10/2025 Guanajuato

CIMAT School of Evolutionary Computation

Selected to participate in the CIMAT School of Evolutionary Computation, an academic program focused on evolutionary computation and computational intelligence. Studied evolutionary algorithms and optimization methods through lectures and practical activities.

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PUBLICATIONS

2026

Genetic Algorithm-Based Optimization of Feature Selection for Arrhythmia Classification in ECG Signals

Manuscript submitted for peer review (decision pending as of September 2026). Compared genetic-algorithm-based feature selection against a full feature set and PCA for binary ECG arrhythmia classification on the MIT-BIH Arrhythmia Database (109,398 heartbeats). The GA-selected 11-feature subset outperformed both baselines across all metrics (91.6% accuracy, F1 macro 0.844), while empirically demonstrating the effect of patient-level data leakage in naive cross-validation protocols (99.1% vs. 90.2% accuracy).

Pending peer review

Authors: Josue Valenzuela Perez | **Journal Name:** Jóvenes en la Ciencia — Congreso Internacional de Electrónica y Cómputo Aplicado (ICEComp) 2026 | **Volume, Issue and Pages:** Vol. 46 (submission under review) | **Publisher:** Universidad de Guanajuato

Link <https://github.com/JosueVP17/ga-ecg-arrhythmia/blob/main/Optimizaci3n de Selecci3n de Características mediante Algoritmo Genético.pdf>

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SKILLS

machine learning | deep learning | principles of artificial intelligence | artificial neural networks | apply statistical analysis techniques | computer vision | C++ | Python (computer programming) | signal processing | perform data analysis | use databases | software frameworks | perform scientific research | analyse scientific data | disseminate results to the scientific community | monitor developments in field of expertise | prepare scientific reports | use data processing techniques

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LANGUAGE SKILLS

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C1	B2	B2	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user